COCOSENSE

AI-Powered Drone Based System for Comprehensive Coconut Tree Health Monitoring and Yield Prediction

Target Users	Innovation Type	Current Status
Coconut Farmers	AI + Drones + IoT + Mobile	Working Multi-Module Prototype

Project Overview

CocoSense is a university-developed smart agriculture solution delivered through a mobile application. It uses drone-captured imagery, deep learning, and IoT-enabled plantation monitoring to help coconut farmers detect pests and diseases early, monitor tree health, estimate yield, and manage plantations more effectively. By turning manual plantation monitoring into a faster, data-driven workflow, CocoSense improves accuracy, timeliness, and decision-making in coconut cultivation.

Problem Addressed

Coconut farmers still depend heavily on manual observation to identify pest attacks, leaf diseases, unhealthy trees, and expected yield. This process is slow, labor-intensive, and highly dependent on human experience. As a result, warning signs are often missed, treatment is delayed, and productivity losses can increase. Traditional yield estimation methods are also unreliable because they do not capture tree-level variation or hidden coconuts within dense canopies.

Core Value Proposition

- Early detection of pests, diseases, and unhealthy trees
- More accurate tree-level yield prediction using drone images
- Multilingual farmer guidance in Sinhala, English, and Tamil
- Plantation-level visibility through GPS mapping and digital records
- A single integrated platform instead of disconnected tools

Who Benefits

- Small and medium-scale coconut farmers
- Estate and plantation managers
- Agricultural field officers
- Coconut development and advisory institutions

Why It Matters

CocoSense helps move coconut farming from reactive manual monitoring to proactive, technology-enabled plantation management. It supports quicker intervention, better productivity planning, and more confident decision-making for both individual trees and full estates.

Integrated Features and Workflow

CocoSense combines multiple AI modules within one mobile-based ecosystem. Each feature solves a specific plantation challenge while contributing to a single, practical decision-support workflow.

Drone Capture	AI Analysis	Detection Results	Guidance & History	Map & Estate Summary

Module	Function	Performance
Health Monitoring	Analyzes leaf and branch images to classify trees as healthy or unhealthy and flag early stress symptoms.	Leaf 93.70% Branch 99.63%
Disease Detection	Identifies major coconut leaf diseases including Leaf Rot, Leaf Spot, and Leaf Dieback to support early diagnosis and treatment planning.	98.69%
Pest Detection	Detects Coconut Mite, Black-Headed Caterpillar, and White Fly and connects the result to multilingual treatment guidance.	Mite 91.44% Unified 96.08%
Yield Prediction	Uses YOLOv8-based coconut detection with dual-view image capture to reduce missed detections and double counting.	87.86%
Bunch Detection	Supports yield estimation by identifying fruit clusters inside the canopy and improving structural understanding of fruit grouping.	88.96%

Added Plantation Intelligence

CocoSense also integrates GPS-based plantation mapping, scan history, and a tree-linked record system. A dedicated IoT-based plantation monitoring device is currently being developed to assign precise locations to each coconut tree, enabling tree-by-tree tracking of health status, detected pests and diseases, and yield-related outputs.

Technology Stack and Measured Impact

Technologies Used

- Drone-based aerial image acquisition from real plantation environments
- EfficientNetB0 for coconut mite detection
- MobileNetV2 for disease, pest, leaf health, and branch health classification
- YOLOv8 for coconut detection, bunch detection, and yield estimation
- Transfer learning, Focal Loss, threshold tuning, and dual-view capture logic
- React Native mobile application with Flask REST API backend
- MongoDB for scan history, user data, and tree records
- GPS / IoT tracking with Google Maps-based plantation visualization
- AI advisory support with multilingual responses

Benefits Delivered

- Early intervention before damage spreads
- Reduced reliance on slow manual inspection
- More consistent disease and pest diagnosis
- Improved harvest planning and labor allocation
- Tree-level and estate-level monitoring in one system
- Multilingual usability for Sri Lankan farmers
- Scalable digital records for smarter plantation management

Performance Highlights

Mite	Unified Pest	Disease	Leaf Health	Branch Health	Bunch	Yield
91.44%	**96.08%**	**98.69%**	**93.70%**	**99.63%**	**88.96%**	**87.86%**

Why This Technology Matters

CocoSense sits at the intersection of Artificial Intelligence, Computer Vision, Precision Agriculture, and IoT-enabled smart farming. It demonstrates how university research can be converted into a practical, scalable product for plantation management and farmer decision support.

Commercial Potential and Future Vision

Commercialization Path

CocoSense can be commercialized as a mobile-based agri-tech solution for coconut farmers, plantations, agricultural cooperatives, and advisory institutions. A practical business model is a tiered subscription structure: a basic plan for individual farmers and an advanced plan for estates requiring dashboards, map-based monitoring, analytics, and structured records.

• Basic mobile scanning and treatment guidance for farmers
• Premium plantation dashboard for estate and enterprise users
• Institutional partnerships with plantation and advisory bodies
• Future hardware-plus-software bundling with drone or IoT services

Future Enhancements

A major next step is the dedicated IoT-based plantation monitoring device currently under development. Its purpose is to assign precise location data to every individual coconut tree in the estate. Once integrated, each tree will be digitally mapped and linked to its own health status, detected pests, diagnosed diseases, and yield prediction outputs.

• Full tree-by-tree plantation mapping and monitoring
• Estate-wide summary output for health, disease, pest, and yield patterns
• Identification of infected zones, unhealthy clusters, and low-yield areas
• Predictive analytics, stronger dashboards, and future offline field inference

Vision Statement

Our vision is to grow CocoSense from a multi-module university innovation into a location-aware digital plantation intelligence platform for coconut farming. By combining AI, drones, mobile computing, multilingual advisory support, and IoT-based mapping, the system aims to improve farmer productivity, enable smarter interventions, and support the digital transformation of Sri Lanka's coconut sector.

Developed as a University Research Innovation

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